Project Proposal: Morphological Profiling of Chemical Compounds Using Cell Painting and High-Content Microscopy

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ABSTRACT

This project proposes a data-driven analysis of drug-induced cellular morphological changes using high-content fluorescence microscopy. We will use the BBBC021 dataset from the Broad Bioimage Benchmark Collection, which contains Cell Painting images of MCF-7 human breast cancer cells treated for 24 hours with a panel of chemical compounds (133 small molecules) at 8 different concentrations and imaged at several time points spanning up to ten weeks after treatment. The dataset can be found in https://bbbc.broadinstitute.org/BBBC021

The objective is to quantitatively assess how drug exposure alters cellular morphology and how these phenotypic changes evolve over time (if time permits). Rather than focusing on predictive performance alone, the project emphasizes biological interpretation of image-derived features, aiming to identify which cellular compartments and morphological descriptors are most affected by chemical perturbations. All analyses will be implemented in Python using image processing and data science techniques and related libraries. This project is conducted within the course *BIO_52101_EP – Sciences des données de l'imagerie biologique*.

1. Introduction

High-content microscopy [1] enables systematic investigation of cellular responses to chemical and genetic perturbations by capturing rich morphological information at single-cell resolution. The Cell Painting assay, in particular, labels multiple cellular compartments (e.g DAPI to image nucleus) using multiplexed fluorescence imaging, allowing subtle phenotypic changes to be quantified and compared across experimental conditions.

The aim of this project is to assess morphological changes induced by drug treatment using the BBBC021 dataset. In addition to comparing treated and control cells, the dataset provides images acquired at multiple time points following treatment, enabling exploration of temporal dynamics in cellular responses. This project aligns with the objectives of the course BIO_52101_EP by combining biomedical imaging, data science methodologies, and biological interpretation.

2. Dataset Description

The BBBC021 dataset from the Broad Bioimage Benchmark Collection consists of fluorescence microscopy images of MCF-7 breast cancer cells treated with a panel of small-molecule compounds. Images were acquired using the Cell Painting protocol, which captures 3 fluorescence channels highlighting cellular structures such as the DNA, F-actin, and B-tubulin.

For each compound, cells were imaged at 8 concentrations and at several time points spanning up to 10 weeks after treatment. This temporal structure enables the analysis of both early and longer-term morphological effects. Depending on exploratory analysis and feasibility constraints, the project may focus on a subset of time points (such as only 2-3 separate weeks) or compounds while preserving biologically meaningful comparisons. For detailed information on the data acquisition process, refer to [2].

3. Biological Objectives and Research Questions

The primary objective of this project is to quantitatively characterize morphological changes induced by drug treatment and to interpret these changes in a biological context. The following research questions may be addressed depending on time availability:

ORCID(s):

- What morphological features distinguish drug-treated cells from untreated controls?
- How do cellular phenotypes evolve over time following drug exposure?
- Which cellular compartments contribute most strongly to observed phenotypic changes (if we do our analysis in all fluorescence channels instead of just DAPI channel)?
- Do compounds with similar mechanisms of action induce similar morphological trajectories?

The emphasis is placed on biological relevance and interpretability rather than on maximizing classification accuracy.

4. Methodology

4.1. Image Processing and Feature Extraction

Raw multi-channel images will be processed using Python-based image analysis tools. Nuclei and whole cells will be segmented using classical image-processing methods (such as Otsu thresholding or using AI methods such as CellSAM [3]. From the segmented cells, quantitative morphological features will be extracted, including intensity, area, shape, texture, and spatial organization descriptors.

4.2. Phenotypic and Temporal Analysis

Single-cell features will be aggregated into treatment-level morphological profiles. Dimensionality reduction methods such as PCA and UMAP will be used to visualize phenotypic similarities and differences across compounds, concentrations, and time points. Clustering or simple classification approaches may be applied to evaluate whether morphological profiles align with known drug mechanisms of action.

4.3. Biological Interpretation

Observed morphological patterns will be interpreted in relation to known biological processes such as cytotoxicity, cell cycle arrest, cytoskeletal reorganization, or mitochondrial stress. Particular attention will be paid to temporal trends, assessing whether phenotypic changes are progressive, stable, or reversible over the course of drug exposure.

5. Expected Outcomes and Course Relevance

This project is expected to produce a biologically interpretable characterization of drug-induced morphological changes and their temporal evolution. The analysis will demonstrate how high-content imaging data can be leveraged to extract meaningful biological insights beyond purely algorithmic performance.

The project fulfills the requirements of the course BIO_52101_EP – Sciences des données de l'imagerie biologique by using a real biomedical imaging dataset, implementing analysis pipelines in Python, and emphasizing biological interpretation and methodological coherence.

References

[1] M. Boutros, F. Heigwer, C. Laufer, Microscopy-based high-content screening, Cell 163 (6) (2015) 1314–1325.

[2] P. D. Caie, R. E. Walls, A. Ingleston-Orme, S. Daya, T. Houslay, R. Eagle, M. E. Roberts, N. O. Carragher, High-content phenotypic profiling of drug response signatures across distinct cancer cells, Molecular cancer therapeutics 9 (6) (2010) 1913–1926.

[3] U. Israel, M. Marks, R. Dilip, Q. Li, C. Yu, E. Laubscher, A. Iqbal, E. Pradhan, A. Ates, M. Abt, et al., Cellsam: a foundation model for cell segmentation, BioRxiv (2025) 2023–11.